

Write a c/c++ program to list all processes in user-mode

Target

1. Write a c/c++ program in kylin desktop os v10
2. To list all processes in user-mode在用户态列出所有进程信息

```
ps -ef
```

类似如下结果: UID, PID, PPID, C, STIME, TTY, TIME, CMD

```
活动 终端 10月19日 17:42 拼
albert@albert-VirtualBox: ~
albert@albert-VirtualBox:~$ ps
  PID TTY          TIME CMD
 2413 pts/0    00:00:00 bash
 2428 pts/0    00:00:00 ps
albert@albert-VirtualBox:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1         0  0 17:35 ?        00:00:01 /sbin/init splash
root           2         0  0 17:35 ?        00:00:00 [kthreadd]
root           3         2  0 17:35 ?        00:00:00 [rcu_gp]
root           4         2  0 17:35 ?        00:00:00 [rcu_par_gp]
root           5         2  0 17:35 ?        00:00:00 [slub_flushwq]
root           6         2  0 17:35 ?        00:00:00 [netns]
root           7         2  0 17:35 ?        00:00:00 [kworker/0:0-events]
root           8         2  0 17:35 ?        00:00:00 [kworker/0:0H-events_highpri]
root          10         2  1 17:35 ?        00:00:06 [kworker/u8:0-flush-8:0]
root          11         2  0 17:35 ?        00:00:00 [mm_percpu_wq]
root          12         2  0 17:35 ?        00:00:00 [rcu_tasks_kthread]
root          13         2  0 17:35 ?        00:00:00 [rcu_tasks_rude_kthread]
root          14         2  0 17:35 ?        00:00:00 [rcu_tasks_trace_kthread]
root          15         2  0 17:35 ?        00:00:00 [ksoftirqd/0]
root          16         2  0 17:35 ?        00:00:00 [rcu_preempt]
root          17         2  0 17:35 ?        00:00:00 [migration/0]
root          18         2  0 17:35 ?        00:00:00 [idle_inject/0]
root          19         2  0 17:35 ?        00:00:00 [rcu_hb/0]
```

1. GCC
2. IDE 集成开发环境
3. in user-mode
 - i. /proc dir contains all informations of all processes

Tools

Install GCC Software Collection

```
sudo apt-get install build-essential
```

How to use GCC

- [gcc and make](#)

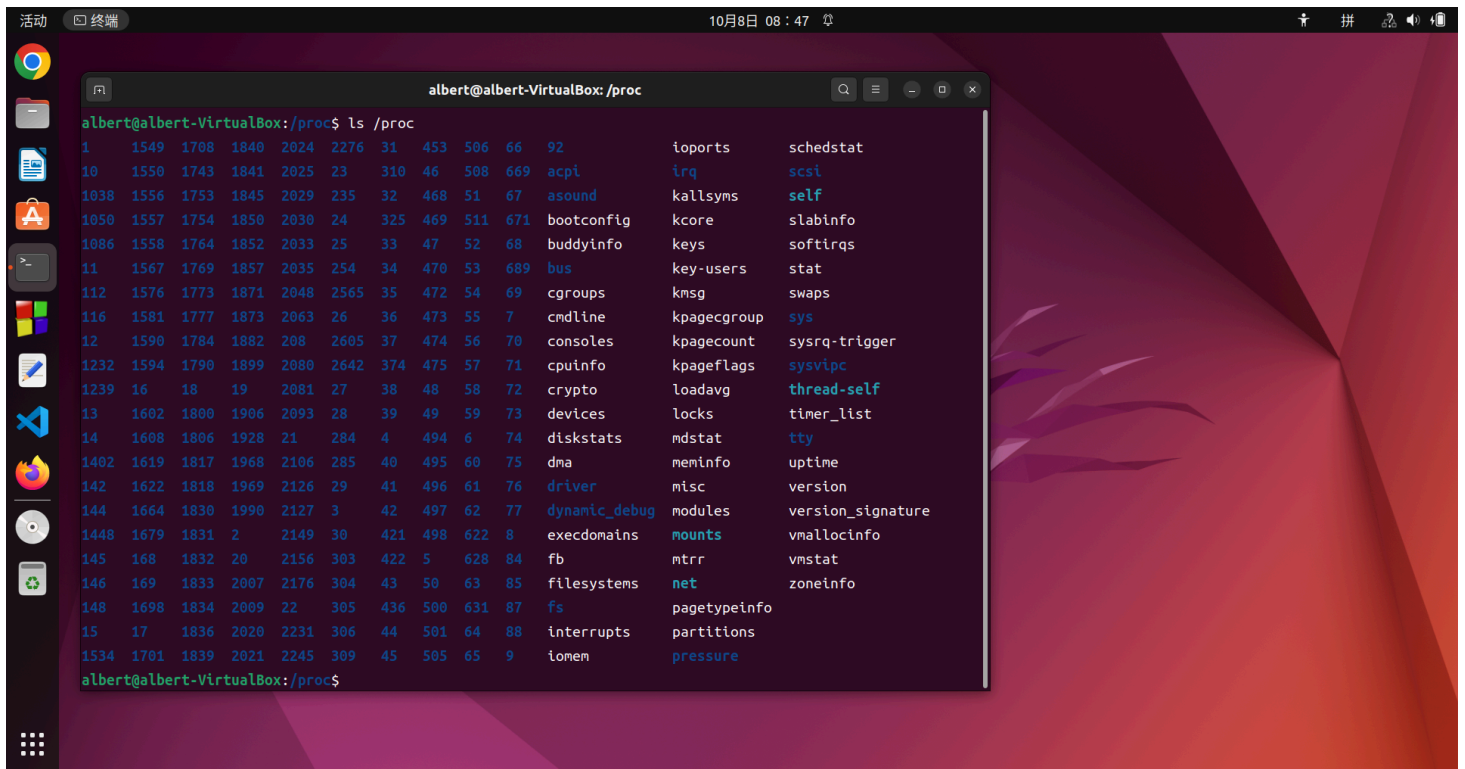
IDE

1. (推荐)Kylin-IDE

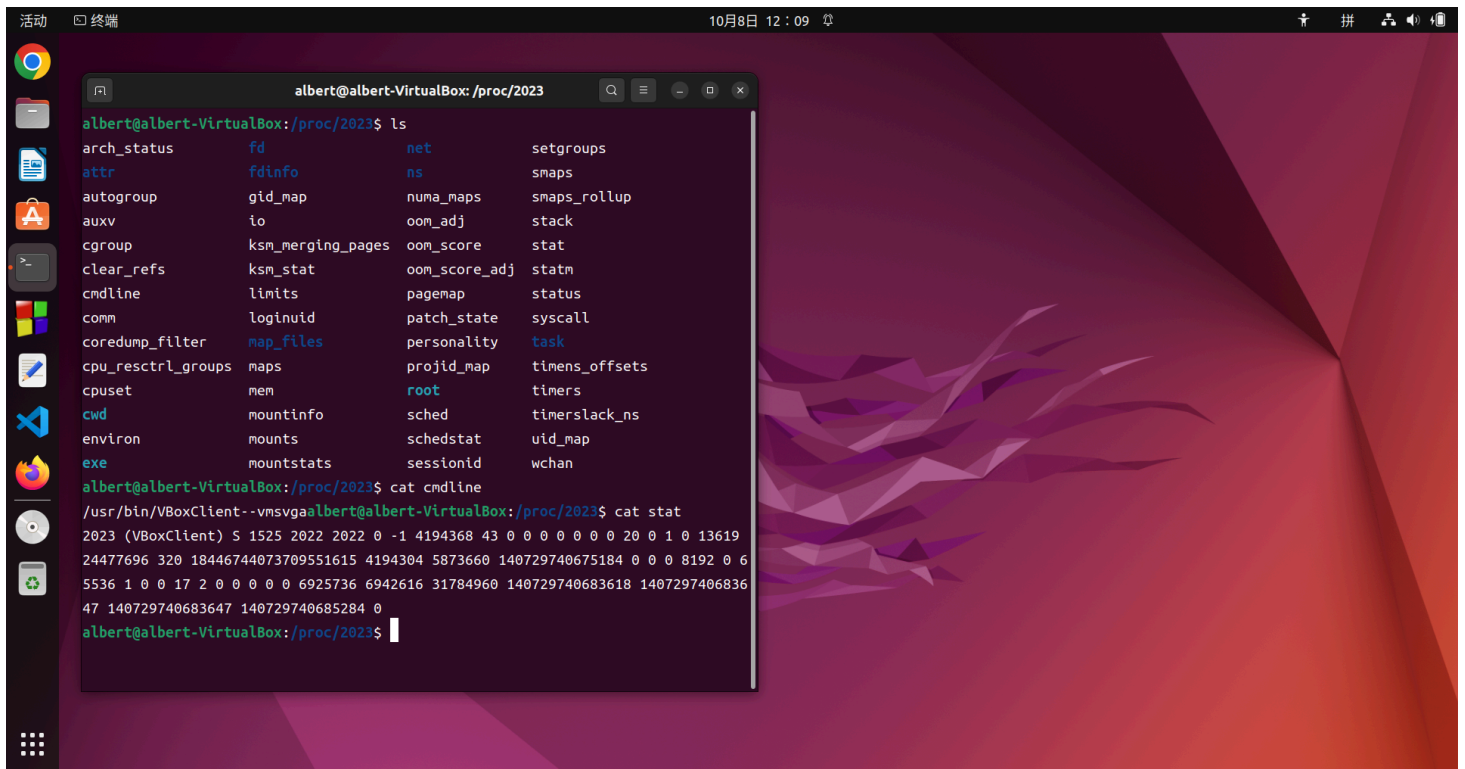
```
sudo apt-get install kylin-ide
```

2. *vim* + *gcc*

/porc



```
cd /proc/2023
```



status file:

```
$ more status
Name:      VBoxClient
Umask:     0027
State:     S (sleeping)
Tgid:      2023
Ngid:      0
Pid:       2023
PPid:      1525
TracerPid: 0
Uid:       1000      1000      1000      1000
Gid:       1000      1000      1000      1000
FDSize:    64
Groups:    4 24 27 30 46 122 134 135 999 1000
NSTgid:    2023
NSpid:     2023
NSpgid:    2022
NSsid:     2022
VmPeak:    23904 kB
VmSize:    23904 kB
VmLck:     0 kB
VmPin:     0 kB
VmHWM:     1280 kB
VmRSS:     1280 kB
RssAnon:   256 kB
...
```

cmdline file:

```
$cat cmdline
usr/sbin/cups-browsed
$
```

stat file:

```
$cat stat
778 (cups-browsed) S 1 778 778 0 -1 4194560 781 0 22 0 2 0 0 0 20 0 3 0 4249 176762880 2816 184
$
```

structure of directory

```
struct dirent
{
    ino_t d_ino; //d_ino 此目录进入点的inode
    off_t d_off; //d_off 目录文件开头至此目录进入点的位移
    signed short int d_reclen; //d_reclen _name 的长度, 不包含NULL 字符
    unsigned char d_type; //d_type d_name 所指的文件类型 d_name 文件名
    char d_name[256];
};
```

the value returned in d_type:

DT_BLK	This is a block device.
DT_CHR	This is a character device.
DT_DIR	This is a directory.
DT_FIFO	This is a named pipe (FIFO).
DT_LNK	This is a symbolic link.
DT_REG	This is a regular file.
DT_SOCK	This is a UNIX domain socket.
DT_UNKNOWN	The file type could not be determined.

```
opendir()
readdir()
closedir()
```

Create a symbol link file

```
#include <fcntl.h>           /* Definition of AT_* constants */
#include <unistd.h>
int link(const char *oldpath, const char *newpath);
```

ps

```
$ ps -ef
```

UID	PID	PPID	C	STIME	TTY	TIME	CMD
root	1	0	0	12:03	?	00:00:01	/sbin/init splash
root	2	0	0	12:03	?	00:00:00	[kthreadd]
root	3	2	0	12:03	?	00:00:00	[rcu_gp]
root	4	2	0	12:03	?	00:00:00	[rcu_par_gp]
root	5	2	0	12:03	?	00:00:00	[slub_flushwq]
root	6	2	0	12:03	?	00:00:00	[netns]
root	7	2	0	12:03	?	00:00:00	[kworker/0:0-events]
root	8	2	0	12:03	?	00:00:00	[kworker/0:0H-events_highpri]
root	9	2	0	12:03	?	00:00:00	[kworker/u8:0-events_unbound]
root	10	2	0	12:03	?	00:00:00	[mm_percpu_wq]
root	11	2	0	12:03	?	00:00:00	[rcu_tasks_kthread]
root	12	2	0	12:03	?	00:00:00	[rcu_tasks_rude_kthread]
...							

How to do

write a c program to list all processes in user-mode, the result is same to ps cmd

1. 列出所有进程，输出结果类似如下命令结果

```
ps -ef
```

...

1. Example of traverse one directory

```
#include <dirent.h>
#include <unistd.h>
#include <stdlib.h>

int main()
{
    DIR * dir;
    struct dirent * ptr;
    /*open dir*/
    dir = opendir("/home");
    /*read dir entry*/
    while((ptr = readdir(dir)) != NULL)
    {
        printf("d_name : %s", ptr->d_name);
        if (ptr->d_type==DT_DIR){
            printf("\tDir");
        }
        printf("\n");
    }
    /*close dir*/
    closedir(dir);
    exit(0);
}
```

Compiling:

```
gcc    listdir.c    -o listdir
./listdir
```